Report 7: Computation Load of the IZak and Zak Transforms

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S7

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October 12, 2026

1 What this report is about

- The slides give operation counts for the transmitter (IZak plus pulse shaping) and the receiver (pulse projection plus Zak). How many operations per frame and per second does that mean, and where is the cost?
- We reproduce the slide's numbers, count the operations of a real FFT and of a complete small transmitter/receiver chain, and ask how much accuracy each extra pulse tap buys.

2 The concept in brief

- **Two blocks on the transmit side.**
  1. The IZak row transform: an N-point IFFT for each of the M delay rows.
  2. Pulse synthesis: every one of the MN values $v_m[q]$ is multiplied into L_g pulse taps and added onto the sample stream.
- **The counts (per frame).**

Item	Direct sum	With FFT
IZak transform	MN^2 MACs	$M\frac{N}{2}\log_2 N$ butterflies
Pulse synthesis	MNL_g MACs	MNL_g MACs
Transmitter total	$MN^2 + MNL_g$	$M\frac{N}{2}\log_2 N + MNL_g$

  (MAC = multiply–accumulate.)
- **Per symbol** (divide by MN): the direct transform costs N operations per symbol, the FFT only $\frac{1}{2}\log_2 N$; the pulse costs L_g per symbol either way.
- **Rates.** Frames arrive at $f_{\text{frame}} = B/MN$, so the load per second is $B\,(N + L_g)$ (direct) or $B\,L_g + \frac{B}{2}\log_2 N$ (FFT). Note that M drops out: the cost depends on the bandwidth, not on the frame size.
- **The receiver mirrors the transmitter.** MNL_g multiplications and $MN(L_g - 1)$ additions for the projection, then the FFT.
- **Role of D.** D does not appear in the transform. It enters only through L_g: a pulse that spans a fixed time in units of T_s has $L_g \approx 2KD + 1$ taps if it spans $\pm K\,T_s$.
- **The slide's FFT count is a bound.** $\frac{N}{2}\log_2 N$ treats every butterfly as having one full multiplication, but twiddle factors such as $1, -1, \pm j$ cost nothing.

One-line memory aid: the FFT makes the transform almost free ($\frac{1}{2}\log_2 N$ per symbol), so the *pulse* (L_g per symbol) is where the cost is. D only matters through L_g, and a longer pulse buys accuracy.

3 What was simulated

- The slide example (M=N=128, B=500 MHz, L_g=31): every number reproduced.
- A radix-2 FFT with counters, for $N = 2, \ldots, 1024$.
- A complete counted transmitter and receiver on a small grid (M=16, N=8, D=4, L_g=33).
- Cost against N and against D, and the accuracy (Nyquist residual ε of Report 4) against L_g for the Gauss×sinc and orthogonalised pulses (M=32, N=16, D=4).

4 Results and discussion

4.1 The slide example

Table 1: Slide example, M=N=128, B=500 MHz, L_g=31. All values match the slides.

Quantity	Value	Per symbol
Symbols per frame MN	16 384	
Direct IZak transform	2 097 152 MACs	128
FFT IZak transform	57 344 butterflies	3.5
Pulse synthesis	507 904 MACs	31
Transmitter total, FFT	565 248	34.5
Transmitter total, direct	2 605 056	159
Frame rate	30 517.6 frames/s	
IZak FFT rate	1.75 G butterflies/s	
Pulse rate	15.5 G MAC/s	
Transmitter rate, direct	79.5 G MAC/s	

- **Expected:** the FFT removes the quadratic cost.
- **Observed:** the transform alone is 36.6× cheaper (2 097 152/57 344). But the whole transmitter is only 4.61× cheaper, because the pulse term does not shrink. The pulse is 89.9 % of the FFT-based transmitter cost.
- The receiver counts for this example are $MNL_g = 507\,904$ projection multiplications and 606 208 additions ($MN(L_g - 1) + MN\log_2 N$).

4.2 How the cost grows with N

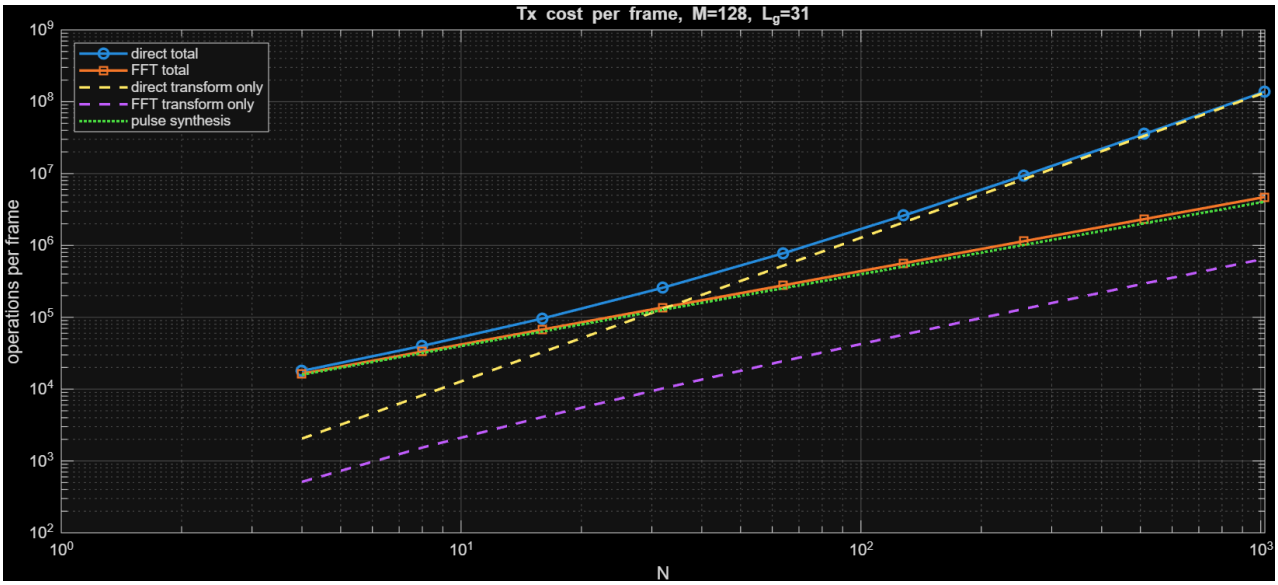


Figure 1: Transmitter cost per frame against N (M=128, L_g=31). Dashed: transform only; dotted: pulse synthesis alone; solid: totals.

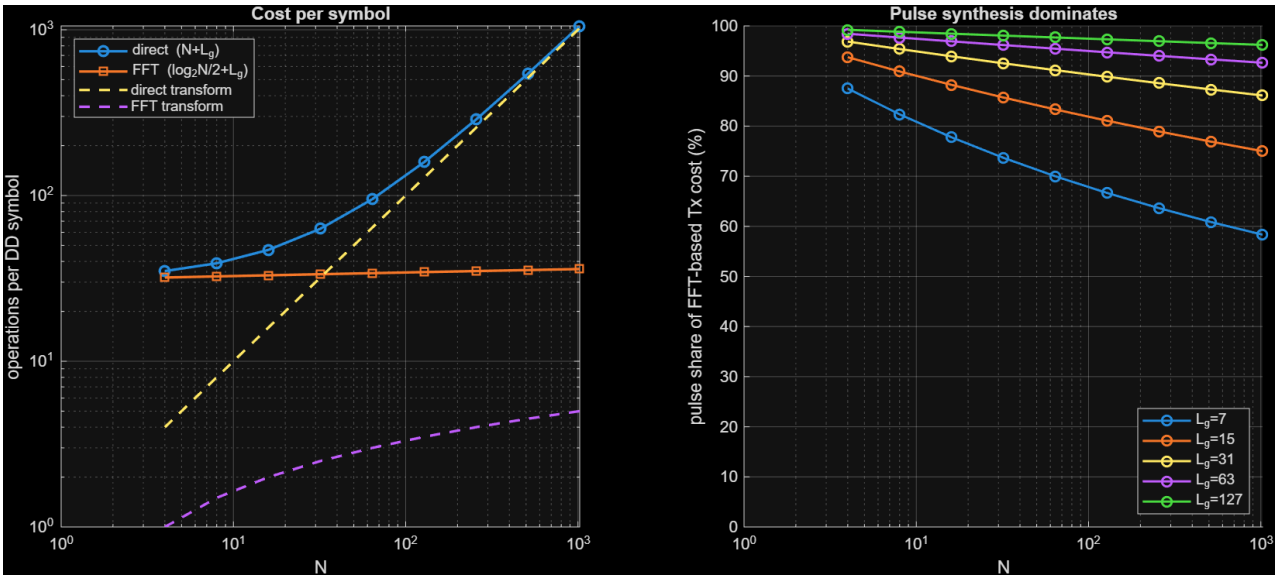


Figure 2: Left: operations per DD symbol. Right: share of the FFT-based transmitter cost that is pulse synthesis, for several L_g.

- **Expected:** the direct transform grows as N^2 per frame (i.e. N per symbol); the FFT as $N \log_2 N$ (i.e. $\frac{1}{2} \log_2 N$ per symbol); the pulse term as N (a constant L_g per symbol).
- **Observed:** for $N \gtrsim 30$ (about where N passes L_g) the direct total climbs away and follows its transform-only line, while the FFT total sits on top of the pulse line: the FFT-based cost is essentially all pulse. The pulse share is 86–90 % for L_g=31 across $N = 128$ to 1024, 96–97 % for L_g=127, and falls to 58 % only for a very short pulse (L_g=7, N=1024).
- Takeaway: once the FFT is used, hardware effort goes into the pulse-shaping filter, not the transform.

4.3 The role of D

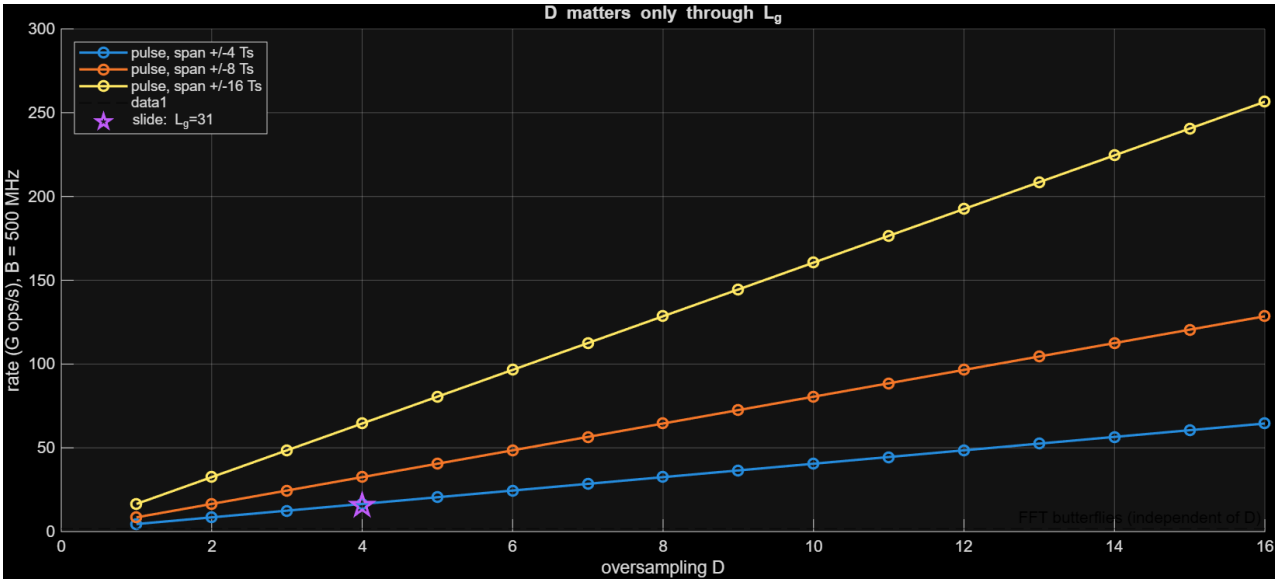


Figure 3: Pulse-synthesis rate ($B\,L_g$, B=500 MHz) against D for pulses spanning ±4, ±8 and $\pm16\,T_s$ ($L_g = 2KD+1$). Star: the slide's L_g=31 at D=4.

- **Expected:** the transform cost does not depend on D; the pulse cost grows linearly with it, at a fixed time span.
- **Observed:** straight lines in D, and the slide's point (L_g=31, 15.5 G MAC/s) lies close to the $\pm4\,T_s$ line (L_g=33, 16.5 G MAC/s at D=4); the slide's pulse spans about $\pm3.75\,T_s$. The FFT rate (1.75 G/s) is independent of D and is tiny on this scale.
- Takeaway: more oversampling costs only through a longer pulse in samples. Doubling D at the same pulse span doubles the dominant cost.

4.4 What each pulse tap buys

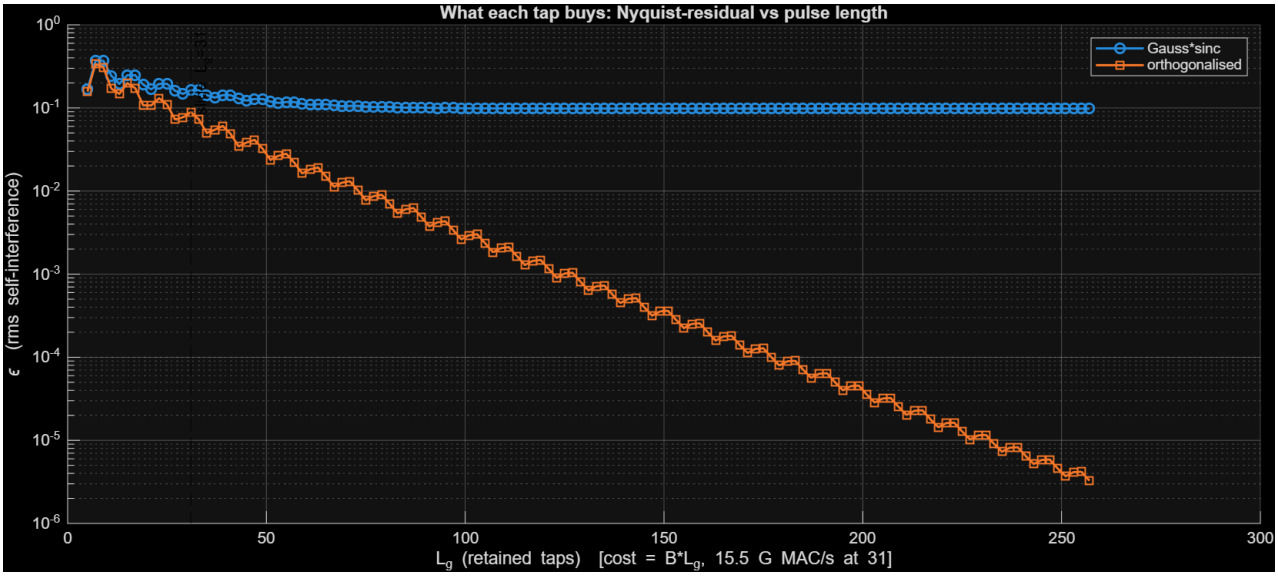


Figure 4: Nyquist residual ε (rms self-interference, Report 4) against the number of retained taps L_g (M=32, N=16, D=4). The cost is BL_g.

Table 2: Accuracy against pulse length, and the pulse rate at B=500 MHz.

L_g	ε Gauss×sinc	ε orthogonalised	pulse rate (G MAC/s)
17	0.247	0.173	8.5
31 (slide)	0.163	0.089	15.5
65	0.111	0.0149	32.5
129	0.098	0.0008	64.5
257	0.098	$\approx 3 \times 10^{-6}$	128.5

- **Expected:** the Gauss×sinc pulse is not orthogonal (Report 4), so longer taps should not help beyond a point; the orthogonalised pulse should keep improving.
- **Observed:** Gauss×sinc levels off near $\varepsilon \approx 0.1$ whatever the length. The orthogonalised pulse improves roughly exponentially with L_g: about a factor 10 for every 45–55 taps.
- **The slide's choice** L_g=31 is therefore a compromise: it leaves an rms self-interference of 9 % (orthogonalised) or 16 % (Gauss×sinc). Reaching $\varepsilon \approx 10^{-3}$ needs about 130 taps, i.e. about 4× the pulse cost (64.5 instead of 15.5 G MAC/s).
- Takeaway: pulse length trades hardware cost against accuracy; the pulse term is the one design knob that matters for computation.

4.5 Is the slide's FFT count too pessimistic?

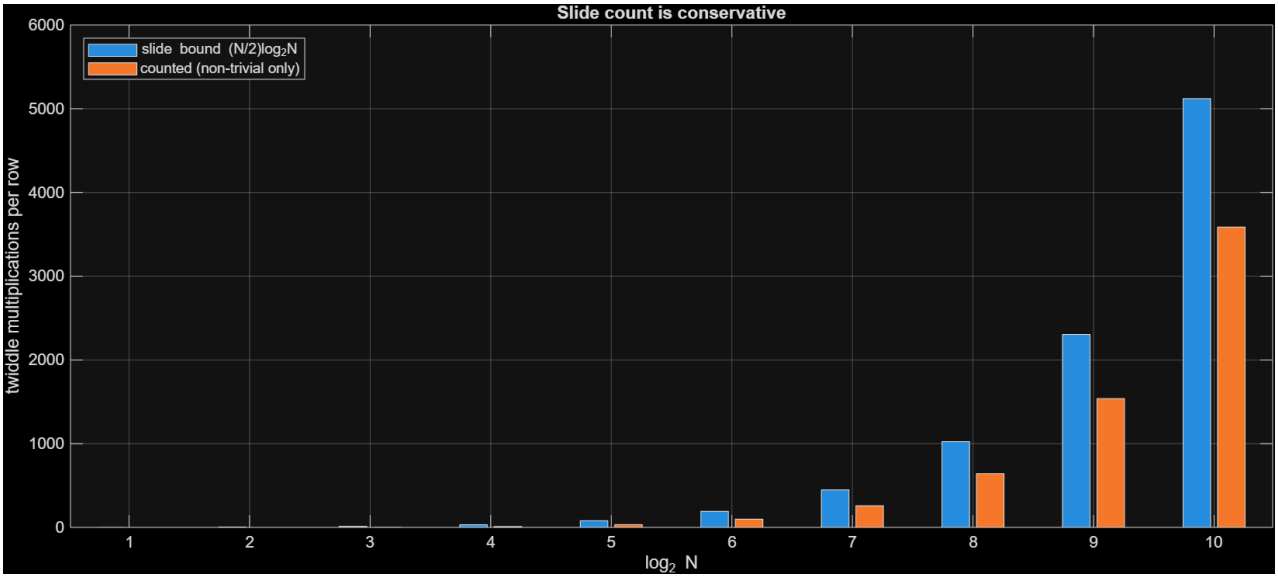


Figure 5: Twiddle multiplications per row: slide bound $\frac{N}{2}\log_2 N$ against the non-trivial ones counted in a real radix-2 FFT.

- **Expected:** the slide itself calls its count conservative, so the true count should be lower.
- **Observed:** the counted non-trivial twiddles equal $\frac{N}{2}\log_2 N - \frac{3N}{2} + 2$ exactly for every $N = 2, \ldots, 1024$ (the counted FFT also matches MATLAB's `fft` to 9×10^{-14} or better, up to $N = 1024$). At N=128: 258 per row, 33 024 per frame, against the slide's 57 344, i.e. 42 % fewer; at N=1024 about 30 % fewer.
- The savings matter little in practice: the FFT is only about 10 % of the transmitter cost.

4.6 A complete counted chain

Table 3: Counted operations in a small chain (M=16, N=8, D=4, L_g=33): counters against formulas.

Item	Counted	Formula
Tx FFT butterflies	192	$M\frac{N}{2}\log_2 N = 192$
Tx pulse MACs	4224	$MNL_g = 4224$
Rx projection multiplications	4224	$MNL_g = 4224$
Rx projection additions	4096	$MN(L_g - 1) = 4096$
Rx FFT butterflies / additions	192 / 384	192 / $MN\log_2 N = 384$

- **Expected and observed:** every counter equals its formula exactly. The counted transmitter and receiver also reproduce the sampled IZak and Zak of Reports 2 and 3 (1.1×10^{-16} and 3.1×10^{-16}).

5 Summary

- The slide's numbers are all reproduced: 2 097 152 direct MACs against 57 344 butterflies, 15.5 G MAC/s for the pulse, 79.5 G MAC/s for the direct transmitter.
- The FFT makes the transform 36.6× cheaper, but the transmitter only 4.6×, because the pulse (89.9 %) dominates.
- D affects the cost only through L_g; the cost depends on B, not on the frame size.
- The Gauss×sinc pulse stalls at $\varepsilon \approx 0.1$ whatever its length; the orthogonalised pulse reaches $\varepsilon \approx 10^{-3}$ at about 130 taps, 4× the cost of the slide's L_g=31.
- The slide's FFT count is a true upper bound: the real count is up to 42 % lower.

Appendix: details and assumptions

- Counts are in complex operations. A complex-times-real MAC (real pulse, as here) needs fewer real multiplications than a complex-times-complex one; the slides make the same remark. We did not convert to real operations.
- The cost model is the one on the slides. Memory traffic, the DAC/ADC chain and any filtering other than the pulse are not counted, and a MAC and a butterfly are counted as one operation each although they are not equally expensive in hardware.
- Trivial twiddles are ±1 and $\pm j$ (the factors with $k = 0$ and, from the second stage on, k = half/2); no other simplification is used.
- The accuracy sweep of Fig. 4 uses M=32, N=16, D=4 (as in Report 6), so its ε values differ from the M=16 values of Report 4. The orthogonalised pulse is the α=16 one of Report 4. The table values are those printed by the script; for the orthogonalised pulse at L_g=257 the script prints 0.00000, so the value $\approx 3\times10^{-6}$ is read from the figure.
- In Fig. 3 the legend contains a meaningless entry ("data1") and the label of the gray dashed line is hard to see; neither affects the content.
- The figures come from `S7_computation_load.m` (MATLAB dark theme).